

M4 · XGBoost (gradient boosting challenger)

Model 5 of the ladder · is LightGBM's result real? · AI Credit Risk / Loan Approval Engine

In one sentence: XGBoost is the same idea as LightGBM (shallow trees built one after another, each fixing the mistakes of all trees before it) with a different engine: trees grow **level by level** up to *max_depth*, where LightGBM grows them leaf by leaf. If two independent libraries agree, the boosting gain is real.

How it was set up

- Same feature sets as M1-M3: **A** 189, **B** 147 features (no LendingClub grades; deployable).
- **Run on Kaggle only** (private CPU notebook, 4 cores, XGBoost 3.4.1, *tree_method=hist*). XGBoost is not installed locally, so a 3-setting smoke test also ran on Kaggle before the full run. Test set still sealed.
- **Random search:** the same 40 settings for A and B at learning rate 0.05 (129 minutes); early stopping on validation AUC after 100 rounds.

Chosen parameters

Parameter	Version A	Version B	What it means
max_depth	3	3	Maximum tree depth (trees grow level by level). Searched 3-10.
min_child_weight	10.23	10.23	Minimum hessian sum per leaf, about loans × 0.14. Searched 1-200.
subsample	0.929	0.929	Share of loans each tree trains on. Searched 0.5-1.0.
colsample_bytree	0.818	0.818	Share of features each tree may use. Searched 0.4-1.0.
reg_lambda	0.002381	0.002381	L2 penalty on leaf weights. Searched 0.001-10.
reg_alpha	7.553	7.553	L1 penalty on leaf weights. Searched 0.0001-10.
gamma	0.761	0.761	Minimum loss reduction needed to split. Searched 0-1.
learning_rate / rounds	0.05 / 2334	0.05 / 2584	Step size and number of trees chosen by early stopping on validation AUC (100-round patience).

Experiments (validation)

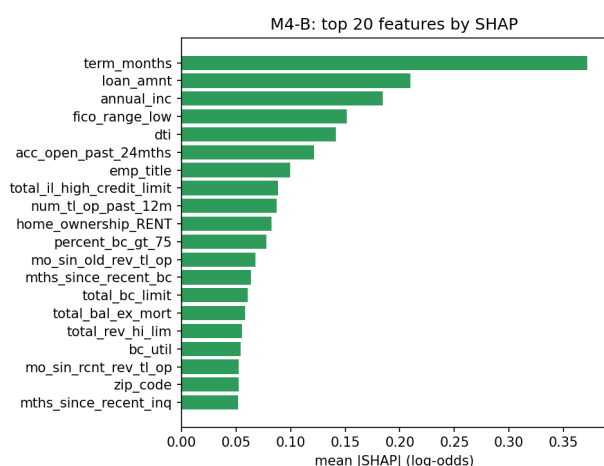
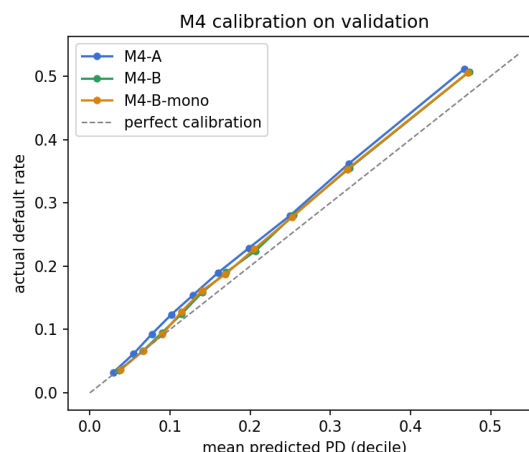
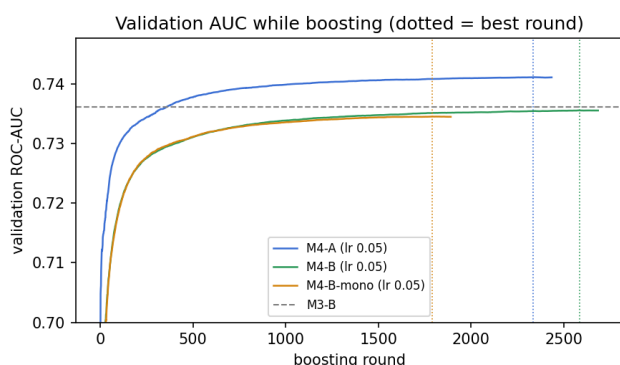
Experiment	Version A	Version B	Takeaway
Default-style baseline → best of search (ROC-AUC)	0.7398 → 0.7412	0.7347 → 0.7359	Tuning adds +0.0012 on B (LightGBM: +0.0009).
Learning rate 0.05 vs 0.02 (ROC-AUC)	0.7412 vs 0.7412	0.7356 vs 0.7356	Kept: 0.05 (A), 0.05 (B); 0.02 only if +0.0005.
scale_pos_weight: ROC-AUC / Brier	0.7405 / 0.2059	0.7350 / 0.2160	Up-weighting defaults again ruins probabilities; not used.
Monotonic constraints (B): ROC-AUC	-	0.7346 (-0.0010)	FICO ↓, DTI ↑, income ↓, 60-month ↑, inquiries ↑, utilisation ↑ risk.

Head-to-head with M3 LightGBM (validation)

Version	M3 ROC-AUC	M4 ROC-AUC	M4 – M3	M3 Brier	M4 Brier	Spearman	Avg of both (info)
A	0.7414	0.7412	-0.0002	0.1422	0.1423	0.992	0.7417
B	0.7362	0.7356	-0.0006	0.1426	0.1427	0.990	0.7365
B-mono	0.7354	0.7346	-0.0009	0.1428	0.1429	0.991	0.7355

Speed on the same 4-core Kaggle CPU: LightGBM 41 s vs XGBoost 97 s per search fit (average). "Avg of both" averages the two models' PDs; shown for information only.

Verdict: a tie that confirms the result. XGBoost and LightGBM are within the 0.005 tie margin on version B (M4 – M3 = -0.0006 ROC-AUC) and rank applicants almost identically (Spearman 0.990). The boosting gain over Logistic Regression is real, not a quirk of one library. By the strategy's rule (keep the faster / simpler one in a tie), **LightGBM** is preferred on speed: about 41 s vs 97 s per search fit on the same Kaggle CPU.



Why one risky applicant got PD 55.1% (average 18.7%; actually defaulted: True)

Feature	Value	SHAP (log-odds)
term_months	60	+0.634
acc_open_past_24mths	12	+0.372
hs		
fico_range_low	660	+0.226
emp_title	0.2329	+0.193
percent_bc_gt_75	11.1	-0.173
mths_since_recent_bc	1	+0.160
inq_last_6mths	4	+0.144

Top-10 SHAP features shared with M3: **9 / 10**. Both libraries rely on the same drivers.

Key findings

- **Version B ROC-AUC:** M1 0.7252 → M3 0.7362 → M4 **0.7356**; Brier M3 0.1426 vs M4 0.1427. Version A: M3 0.7414 vs M4 0.7412.
- **Approval view (B):** approving the safest 50% gives a bad rate of 9.57% (M4) vs 9.56% (M3) and 10.03% (M1).
- **Monotonic constraints** cost +0.0010 ROC-AUC with XGBoost (M3: +0.0008): nearly free in both libraries.
- **Overfitting:** train ROC-AUC 0.7471 vs validation 0.7356 for B. **Calibration:** mean PD 18.74% vs actual 20.35%, so the final model still needs calibrating on validation.
- **Segments (B):** ROC-AUC 0.700 (36-month) vs 0.685 (60-month). **LendingClub's grades:** A beats B by 0.0056 ROC-AUC.